

U-Statistics

Econometric Methods for Social Spillovers and Networks

Universität St.Gallen, September 21st to 25th, 2026

Bryan S. Graham

University of California - Berkeley

U-Statistics

Introduced by Wassily Hoeffding (1948).

Arise frequently in semiparametric econometrics.

Also useful for the analysis of dyadic data (and network data more generally).

References: Lehmann (1999, Chapter 6), van der Vaart (2000, Chapter 12) and Ferguson (2005).

A familiar example

Let

$$\bar{X}_N = \frac{1}{N} \sum_{i=1}^N X_i$$

be the sample mean and

$$S^2 = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2$$

the sample variance.

It turns out that S^2 is a second order U-statistic.

Sample variance

$$\begin{aligned} S^2 &= \frac{1}{2N(N-1)} \sum_{i=1}^N \sum_{j=1}^N \left((X_i - \bar{X}_N)^2 + (X_j - \bar{X}_N)^2 \right) \\ &= \frac{1}{2N(N-1)} \sum_{i=1}^N \sum_{j=1}^N \left((X_i - \bar{X}_N) - (X_j - \bar{X}_N) \right)^2 \\ &= \frac{1}{N(N-1)} \sum_{i=1}^N \sum_{j=1}^N \frac{1}{2} (X_i - X_j)^2 \\ &= \frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=1+1}^N \frac{1}{2} (X_i - X_j)^2 \\ &= \binom{N}{2}^{-1} \sum_{i < j} \frac{1}{2} (X_i - X_j)^2. \end{aligned}$$

Unbiasedness under random sampling

If $\{X_i\}_{i=1}^N$ are i.i.d random draws from F_Z , then $\mathbb{E}[S^2]$ is unbiased for $\mathbb{V}(X_9)$:

$$\begin{aligned}\mathbb{E}[S^2] &= \frac{1}{2}\mathbb{E}[(X_9 - X_{19})^2] \\ &= \frac{1}{2}\mathbb{E}[(X_9 - \mathbb{E}[X_9] - (X_{19} - \mathbb{E}[X_{19}]))^2] \\ &= \frac{1}{2}\mathbb{E}[(X_9 - \mathbb{E}[X_9])^2 + (X_{19} - \mathbb{E}[X_{19}])^2] \\ &= \mathbb{E}[(X_9 - \mathbb{E}[X_9])^2] \\ &= \mathbb{V}(X_9)\end{aligned}$$

(X_9 and X_{19} are my favorite draws under random sampling when when $N \geq 19$).

Definition: U-Statistic

Let $\{X_i\}_{i=1}^N$ be a simple random sample from F_X .

Let $h(X_{i_1}, \dots, X_{i_m})$ be a symmetric *kernel* function.

(We can always replace $h(X_{i_1}, \dots, X_{i_m})$ with its average across permutations – resulting in symmetry).

A U-statistic is an average of the kernel $h(X_{i_1}, \dots, X_{i_m})$ over all possible m -tuples of observations in the sample.

$$U_N = \binom{N}{m}^{-1} \sum_{\mathbf{i} \in C_{m,N}} h(X_{i_1}, \dots, X_{i_m})$$

where $C_{m,N}$ denotes the set of all unique combinations of indices of size m drawn from the set $\{1, 2, \dots, N\}$.

Definition: U-Statistic (continued)

The parameter of interest is

$$\theta = \mathbb{E} [U_N] = \mathbb{E} [h (X_1, \dots, X_m)] ,$$

where the expectation is over m independent random draws from the target population.

Our goals for today:

1. U_N is unbiased for θ , what is $\mathbb{V} (U_N)$?
2. large sample theory / asymptotic normality;
3. extension to M-estimation type problems.

Kendall's Tau

$\{(X_i, Y_i)\}_{i=1}^N$ are i.i.d random draws from $F_{X,Y}$.

The probability that a pair of observations are *concordant* is

$$c = \Pr(X_i > X_j \cap Y_i > Y_j) \cup \Pr(X_i < X_j \cap Y_i < Y_j)$$

Kendall's Tau is the population proportion of concordant pairs minus the population proportion of *discordant* pairs

$$\begin{aligned}\tau &= c - (1 - c) \\ &= 2c - 1.\end{aligned}$$

A “nonparametric” measure of correlation/monotonicity: -1 (never concordant) and 1 (always concordant).

Kendall's Tau (continued)

Let $Z_i = (X_i, Y_i)$ and define the kernel $h(Z_i, Z_j)$ as

$$h(Z_i, Z_j) = 2 \left[\mathbf{1}(X_i > X_j) \mathbf{1}(Y_i > Y_j) + \mathbf{1}(X_i < X_j) \mathbf{1}(Y_i < Y_j) \right] - 1.$$

With some work it is possible to show that

$$h(Z_i, Z_j) = \left(1 - 2\mathbf{1}(X_i < X_j)\right) \left(1 - 2\mathbf{1}(Y_i < Y_j)\right),$$

which will be useful later.

An unbiased estimate of τ is

$$\hat{\tau} = \binom{N}{2}^{-1} \sum_{i < j} h(Z_i, Z_j).$$

(we will symmetrize $h(Z_i, Z_j)$ later).

Variance

The variance of U_N is

$$\begin{aligned}\mathbb{V}(U_N) &= \mathbb{V} \left(\binom{N}{m}^{-1} \sum_{\mathbf{i} \in C_{m,N}} h(X_{i_1}, \dots, X_{i_m}) \right) \\ &= \binom{N}{m}^{-2} \sum_{\mathbf{i} \in C_{m,N}} \sum_{\mathbf{j} \in C_{m,N}} \mathbb{C} \left(h(X_{i_1}, \dots, X_{i_m}), h(X_{j_1}, \dots, X_{j_m}) \right).\end{aligned}\tag{1}$$

(Some of) the summands in $\mathbb{V}(U_N)$ covary.

Fortunately this dependence is structured.

Variance (continued)

For $s = 1, \dots, m$ let

$$\bar{h}_s(x_1, \dots, x_s) = \mathbb{E}[h(x_1, \dots, x_s, X_{s+1}, \dots, X_m)]$$

be the average over the last $m - s$ elements of $h(\cdot)$ holding the first s elements fixed.

Note that since X_{i_k} is independent of X_{i_l} for all $k \neq l$ we have

$$\begin{aligned} \mathbb{E}[h(X_1, \dots, X_s, X_{s+1}, \dots, X_m) | (X_1, \dots, X_s) = (x_1, \dots, x_s)] \\ = \mathbb{E}[h(x_1, \dots, x_s, X_{s+1}, \dots, X_m)]. \end{aligned}$$

It is also useful to observe that

$$\mathbb{E}[\bar{h}_s(X_1, \dots, X_s)] = \mathbb{E}[h(X_1, \dots, X_m)] = \theta.$$

Variance (continued)

Define, for $s = 1, \dots, m$,

$$\delta_s^2 = \mathbb{V} \left(\bar{h}_s (X_1, \dots, X_s) \right).$$

The form of the covariances in (1) depends on the number of indices in common.

Let s be the number of indices in common in $X_{i_1}, \dots, X_{i_m}$ and $X_{j_1}, \dots, X_{j_m}$:

$$\begin{aligned} & \mathbb{C} \left(h \left(X_{i_1}, \dots, X_{i_m} \right), h \left(X_{j_1}, \dots, X_{j_m} \right) \right) \\ &= \mathbb{E} \left[\left(h \left(X_{i_1}, \dots, X_{i_s}, X_{i_{s+1}}, \dots, X_{i_m} \right) - \theta \right) \right. \\ & \quad \left. \times \left(h \left(X_{i_1}, \dots, X_{i_s}, X_{j_{s+1}}, \dots, X_{j_m} \right) - \theta \right) \right] \quad (2) \end{aligned}$$

Variance (continued)

Conditional on $X_1, \dots, X_s$ the two terms in (2) are independent so that, using the Law of Iterated Expectations,

$$\begin{aligned} \mathbb{C} \left(h \left(X_{i_1}, \dots, X_{i_m} \right), h \left(X_{j_1}, \dots, X_{j_m} \right) \right) \\ = \mathbb{E} \left[\left(\bar{h}_s (X_1, \dots, X_s) - \theta \right) \left(\bar{h}_s (X_1, \dots, X_s) - \theta \right) \right] \\ = \delta_s^2. \end{aligned}$$

Using the same argument yields

$$\mathbb{C} \left(\bar{h}_s (X_1, \dots, X_s), h (X_1, \dots, X_m) \right) = \delta_s^2.$$

Variance (continued)

By the Cauchy-Schwartz Inequality we have

$$\frac{\mathbb{C} \left(\bar{h}_s (X_1, \dots, X_s), h (X_1, \dots, X_m) \right)}{\delta_s \delta_m} \leq 1$$

and hence

$$\delta_s^2 \leq \delta_m^2.$$

Continuing with this type of reasoning we get the weak ordering

$$\delta_1^2 \leq \delta_2^2 \leq \dots \leq \delta_m^2.$$

In what follows we will assume that $\delta_m^2 < \infty$.

Variance (continued)

To use these results to get an expression for $\mathbb{V}(U_N)$ begin by observing that the number of pairs of m -tuples $(i_1, \dots, i_m)$ and $(j_1, \dots, j_m)$ having exactly s elements in common is

$$\binom{N}{m} \binom{m}{s} \binom{N-m}{m-s}.$$

Variance (continued)

This follows since:

1. $\binom{N}{m}$ equals the number of ways of choosing $(i_1, \dots, i_m)$ from the set $\{1, \dots, N\}$.
2. For each unique m -tuple there are $\binom{m}{s}$ ways of choosing a subset of size s from it.
3. Having fixed the s indices in common there are then $\binom{N-m}{m-s}$ ways of choosing the $m - s$ non-common elements of $(j_1, \dots, j_m)$ from the $N - m$ integers not already present in $(i_1, \dots, i_m)$.

Variance (continued)

We therefore have

$$\begin{aligned}\mathbb{V}(U_N) &= \binom{N}{m}^{-2} \sum_{s=0}^m \binom{N}{m} \binom{m}{s} \binom{N-m}{m-s} \delta_s^2 \\ &= \sum_{s=1}^m \left[\frac{m!^2}{s! (m-s)!^2} \times \right. \end{aligned} \tag{3}$$

$$\left. \frac{(N-m)(N-m-1) \cdots (N-2m+s+1)}{N(N-1) \cdots (N-m+1)} \right] \delta_s^2 \tag{4}$$

Variance (continued)

To understand this expression note that each of the covariances in (4) above have $s = 0, \dots, m$ elements in common.

The coefficients on the δ_s^2 in (4) give the number of covariances with s elements in common.

Also note that $\delta_0^2 = 0$.

Variance (continued)

The coefficient on δ_1^2 is

$$\begin{aligned} & \frac{m!^2}{1! (m-1)!^2} \frac{(N-m)(N-m-1) \cdots (N-2m+1+1)}{N(N-1) \cdots (N-m+1)} \\ &= m^2 \frac{\overbrace{(N-m)(N-m-1) \cdots (N-2m+2)}^{\text{m-1 terms}}}{\underbrace{N(N-1) \cdots (N-m+1)}_{\text{m terms}}} \end{aligned}$$

$$\approx \frac{m^2}{N}.$$

Variance (continued)

The coefficient on δ_2^2 is $O(N^{-2})$ etc. We therefore have

$$\mathbb{E}[U_N] = \theta, \quad \mathbb{V}(U_N) = \frac{m^2}{N} \delta_1^2 + O(N^{-2})$$

and also that $\mathbb{V}(\sqrt{N}(U_N - \theta)) \rightarrow m^2 \delta_1^2$ as $N \rightarrow \infty$.

If $\delta_1 = 0$ we say that U_N is a degenerate U-Statistic with degeneracy of order 1.

Large sample theory

Basic idea:

1. We are interested in the asymptotic distribution of U_N (*a priori* complicated).
2. Find another statistics U_N^* with well-understood asymptotic distribution.
3. Show that U_N is “close enough” to U_N^* as $N \rightarrow \infty$ such that they have the same asymptotic distribution.

Hajek Projection

The asymptotic properties of sums of independent random variables, appropriately scaled, are especially well-understood.

Let $X_1, X_2, \dots, X_N$ be independent $K \times 1$ random vectors. Let $\mathcal{L}$ be the linear subspace containing of all functions of the form

$$\sum_{i=1}^N g_i(X_i) \tag{5}$$

for $g_i : \mathbb{R}^K \rightarrow \mathbb{R}$ arbitrary with $\mathbb{E} [g_i(X_i)^2] < \infty$ for $i = 1, \dots, N$.

Hajek Projection (continued)

Next let Y be an arbitrary random variable with finite variance, but unknown distribution.

Use the Projection Theorem to approximate the statistic Y with one composed of a sum of independent random functions.

Such a sum, by appeal to a CLT, may be well-described by a normal distribution.

If the projection is also a very good approximation of Y , then the hope is that Y may be accurately described by a normal distribution as well.

Hajek Projection (continued)

The projection of Y onto $\mathcal{L}$, equals

$$\Pi(Y|\mathcal{L}) = \sum_{i=1}^N \mathbb{E}[Y|X_i] - (N-1)\mathbb{E}[Y]. \quad (6)$$

To verify (6) it suffices to check the necessary and sufficient orthogonality condition of the Projection Theorem.

Hajek Projection: verification

It is helpful to observe that, for $j \neq i$,

$$\begin{aligned}\mathbb{E} \left[\mathbb{E} [Y | X_i] | X_j \right] &= \mathbb{E} [\mathbb{E} [Y | X_i]] \\ &= \mathbb{E} [Y],\end{aligned}\tag{7}$$

due to independence of X_i and X_j and the law of iterated expectations.

In contrast, if $j = i$, then

$$\mathbb{E} [\mathbb{E} [Y | X_i] | X_i] = \mathbb{E} [Y | X_i].\tag{8}$$

Hajek Projection: verification (continued)

The orthogonality condition to verify, for $U = Y - \Pi(Y|\mathcal{L})$, is

$$\begin{aligned} 0 &= \mathbb{E} \left[U \left(\sum_{j=1}^N g_j(X_j) \right) \right] \\ &= \sum_{j=1}^N \mathbb{E} \left[\mathbb{E}[U|X_j] g_j(X_j) \right] \end{aligned}$$

Next observe that, using (7) and (8),

$$\begin{aligned} \mathbb{E}[U|X_j] &= \mathbb{E}[Y|X_j] - \sum_{i=1}^N \mathbb{E}[\mathbb{E}[Y|X_i]|X_j] + (N-1)\mathbb{E}[Y] \\ &= \mathbb{E}[Y|X_j] - \mathbb{E}[Y|X_j] - (N-1)\mathbb{E}[Y] + (N-1)\mathbb{E}[Y] \\ &= 0, \end{aligned}$$

for $j = 1, \dots, N$.

Hajek Projection: iid simplification

If, in addition to independence, we have that (i) $\{X_i\}_{i=1}^N$ are identically distributed and (ii) $Y = h(X_1, \dots, X_N)$ is a permutation symmetric function of $\{X_i\}_{i=1}^N$, then

$$\begin{aligned}\mathbb{E}[Y | X_i = x] &= \mathbb{E}[Y | X_1 = x] \\ &= \mathbb{E}[h(x, X_2, \dots, X_N)] \\ &\stackrel{def}{=} \bar{h}_1(x)\end{aligned}$$

for all $i = 1, \dots, N$. Since $\bar{h}_1(x)$ does not depend on i it follows that (6) simplifies, in this case, to

$$\boxed{\Pi(Y | \mathcal{L}) = \sum_{i=1}^N \bar{h}_1(X_i) - (N-1) \mathbb{E}[Y]} \quad (9)$$

Large Sample Theory

Let $\{Y_N\}$ be a sequence of statistics indexed by the sample size and $\mathcal{L}_N$ a corresponding sequence of linear subspaces of form (5).

Goal: use the limiting distribution of $\sqrt{N} (\Pi(Y_N | \mathcal{L}_N) - \Pi(Y_N | 1))$ to approximate that of $\sqrt{N} (Y_N - \Pi(Y_N | 1))$.

Valid if these two statistics converge in mean square (to one another).

Attractive because in many cases of interest the asymptotic sampling distribution of $\sqrt{N} (\Pi(Y_N | \mathcal{L}_N) - \Pi(Y_N | 1))$ is straightforward to derive, whereas that of $\sqrt{N} (Y_N - \Pi(Y_N | 1))$ may be ex ante non-obvious.

Large Sample Theory (continued)

The “Analysis of Variance” decomposition for projections gives

$$\begin{aligned}\|Y_N - \Pi(Y_N | 1)\|^2 &= \|Y_N - \Pi(Y_N | \mathcal{L}_N)\|^2 \\ &\quad + \|\Pi(Y_N | \mathcal{L}_N) - \Pi(Y_N | 1)\|^2,\end{aligned}$$

which, after some re-arrangement, yields

$$\begin{aligned}N \|Y_N - \Pi(Y_N | \mathcal{L}_N)\|^2 &= N \|Y_N - \Pi(Y_N | 1)\|^2 \\ &\quad - N \|\Pi(Y_N | \mathcal{L}_N) - \Pi(Y_N | 1)\|^2.\end{aligned}$$

Or, invoking the covariance inner product, that $\Pi(Y | 1) = \mathbb{E}[Y]$, as well as the definition of variance,

$$N\mathbb{E}[(Y_N - \Pi(Y_N | \mathcal{L}_N))^2] = N\mathbb{V}(Y_N) - N\mathbb{V}(\Pi(Y_N | \mathcal{L}_N)). \quad (10)$$

Large Sample Theory

If the limits of $N\mathbb{V}(Y_N)$ and $N\mathbb{V}(\Pi(Y_N|\mathcal{L}_N))$ coincide as $N \rightarrow \infty$ we have that $\sqrt{N}(Y_N - \Pi(Y_N|\mathcal{L}_N))$ converges in mean square to zero.

This means that $\sqrt{N}Y_N$ and $\sqrt{N}\Pi(Y_N|\mathcal{L}_N)$ will have identical limit distributions.

Application to U-Statistics

The Hajek Projection of U_N onto $\mathcal{L}$ equals, from (6) above,

$$\Pi(U_N | \mathcal{L}_N) = \sum_{i=1}^N \mathbb{E}[U_N | X_i] - (N-1) \mathbb{E}[U_N]. \quad (11)$$

To simplify the argument assume that $m = 2$. The L^2 projection of U_N onto just the first observation X_1 is

$$\begin{aligned} \mathbb{E}[U_N | X_1] &= \binom{N}{2}^{-1} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \mathbb{E}[h(X_i, X_j) | X_1] \\ &= \binom{N}{2}^{-1} (N-1) \bar{h}_1(X_1) + \binom{N}{2}^{-1} \left(\binom{N}{2} - (N-1) \right) \theta \\ &= \frac{2}{N} \{ \bar{h}_1(X_1) - \theta \} + \theta. \end{aligned} \quad (12)$$

Application to U-Statistics (continued)

The second equality follows because $\mathbb{E} \left[h \left(X_i, X_j \right) \middle| X_1 \right] = \bar{h}_1 \left(X_1 \right)$ if either i or j equals 1 (which occurs $N - 1$ times).

In all other cases, by random sampling, $\mathbb{E} \left[h \left(X_i, X_j \right) \middle| X_1 \right] = \mathbb{E} \left[h \left(X_i, X_j \right) \right] = \theta$ (which occurs $\binom{N}{2} - (N - 1)$ times).

Substituting (12) into (11) yields

$$\Pi \left(U_N - \theta \middle| \mathcal{L}_N \right) = \frac{2}{N} \sum_{i=1}^N \left\{ \bar{h}_1 \left(X_i \right) - \theta \right\}.$$

For the general $m \geq 2$ case a similar calculation gives

$$\Pi \left(U_N - \theta \middle| \mathcal{L}_N \right) = \frac{m}{N} \sum_{i=1}^N \left\{ \bar{h}_1 \left(X_i \right) - \theta \right\}.$$

Application to U-Statistics (continued)

Since $\Pi(U_N - \theta | \mathcal{L}_N)$ is a sum of i.i.d. random variables with $\mathbb{V}(\bar{h}_1(X_1) - \theta) = \delta_1^2$, a CLT gives

$$\sqrt{N}\Pi(U_N - \theta | \mathcal{L}_N) \xrightarrow{D} \mathcal{N}(0, m^2\delta_1^2).$$

Our (combinatoric) variance calculations gave

$$\mathbb{V}(\sqrt{N}(U_N - \theta)) \rightarrow m^2\delta_1^2$$

as $N \rightarrow \infty$.

Application to U-Statistics (continued)

Therefore

$$N\mathbb{V}(U_N) - N\mathbb{V}(\Pi(U_N | \mathcal{L}_N)) \rightarrow 0$$

as $N \rightarrow \infty$, in turn implying that $\sqrt{N}(U_N - \theta)$ converges in mean square to $\sqrt{N}\Pi(U_N - \theta | \mathcal{L}_N)$ and hence that

$$\boxed{\sqrt{N}(U_N - \theta) \xrightarrow{D} \mathcal{N}(0, m^2 \delta_1^2)}$$

as needed.

Limit Theory for Kendall's Tau

First we symmetrize the kernel:

$$h(Z_i, Z_j) = \frac{1}{2} \left[\left(1 - 2\mathbf{1}(X_i < X_j) \right) \left(1 - 2\mathbf{1}(Y_i < Y_j) \right) \right. \\ \left. + \left(1 - 2\mathbf{1}(X_j < X_i) \right) \left(1 - 2\mathbf{1}(Y_j < Y_i) \right) \right].$$

Limit Theory for Kendall's Tau (continued)

To calculate the projection we evaluate (under no-concordance H_0)

$$\begin{aligned}\mathbb{E}[h(Z_1, Z_2) | Z_1] &= \frac{1}{2} (1 - 2[1 - F_X(X_1)]) (1 - 2[1 - F_Y(Y_1)]) \\ &\quad + \frac{1}{2} (1 - 2F_X(X_1)) (1 - 2F_Y(Y_1)) \\ &= \frac{1}{2} (2F_X(X_1) - 1) (2F_Y(Y_1) - 1) \\ &\quad + \frac{1}{2} (1 - 2F_X(X_1)) (1 - 2F_Y(Y_1)).\end{aligned}$$

Limit Theory for Kendall's Tau (continued)

Next observe that $U_1 \stackrel{d}{=} 1 - 2F_X(X_1)$ and $V_1 \stackrel{d}{=} 1 - 2F_Y(Y_1)$ with U_1 and V_1 uniform on $[-1, 1]$.

We therefore have

$$\mathbb{E}[h(Z_1, Z_2) | Z_1] = \bar{h}_1(Z_1) \stackrel{d}{=} U_1 V_1.$$

Under the null of independence of X_1 and Y_1

$$\delta_1^2 = \mathbb{V}(U_1 V_1) \stackrel{H_0}{=} \frac{1}{9}$$

(If $U \sim \text{Uniform}[a, b]$, then $\mathbb{E}[U] = (a + b)/2$ and $\mathbb{V}(U) = (b - a)^2/12$).

Limit Theory for Kendall's Tau (continued)

Putting things together we get

$$\boxed{\sqrt{N} (\hat{\tau} - \tau) \xrightarrow[H_0]{D} \mathcal{N} \left(0, \frac{4}{9} \right) .}$$

No need to calculate a variance to test the null of no-concordance.

Exercise: Derive limit theory for “general” case.

U-Process Minimizers

Honoré & Powell (1994) study the large sample properties of U-Process minimizers.

Let $\{Z_i\}_{i=1}^N$ be a sample of i.i.d random variables and consider the estimator $\hat{\beta}$ which minimizes

$$L_N(\beta) = \binom{N}{m}^{-1} \sum_{\mathbf{i} \in C_{m,N}} l(Z_{i_1}, \dots, Z_{i_m}; \beta).$$

A mean value expansion gives, after some manipulation

$$\sqrt{N}(\hat{\beta} - \beta_0) = -\Gamma_0^{-1} \sqrt{N} \left[\binom{N}{m}^{-1} \sum_{\mathbf{i} \in C_{m,N}} \nabla_{\beta} l(Z_{i_1}, \dots, Z_{i_m}; \beta_0) \right] + o_p(1)$$

where $\text{plim}_{N \rightarrow \infty} \nabla_{\beta\beta} L_N(\hat{\beta}) = \Gamma_0$, assumed invertible.

U-Process Minimizers (continued)

To make connections to the basic theory of U-Statistics outlined above define

$$h \left(Z_{i_1}, \dots, Z_{i_m}; \beta \right) = \nabla_{\beta} l \left(Z_{i_1}, \dots, Z_{i_m}; \beta \right)$$

and also

$$\tilde{h}_1 (z_1; \beta) = \mathbb{E} \left[h \left(z_1, Z_{i_2}, \dots, Z_{i_m}; \beta \right) \right] .$$

A CLT gives

$$\frac{m}{\sqrt{N}} \sum_{i=1}^N \tilde{h}_1 (Z_i; \beta_0) \xrightarrow{D} \mathcal{N} \left(0, m^2 \Omega_0 \right) .$$

with

$$\Omega_0 = \mathbb{E} \left[\tilde{h}_1 (Z_i; \beta_0) \tilde{h}_1 (Z_i; \beta_0)' \right] .$$

U-Process Minimizers (continued)

Define

$$U_N(\beta_0) = \binom{N}{m}^{-1} \sum_{\mathbf{i} \in C_{m,N}} \nabla_{\beta} l(Z_{i_1}, \dots, Z_{i_m}; \beta_0),$$
$$U_N^*(\beta_0) = \frac{m}{N} \sum_{i=1}^N \tilde{h}_1(Z_i; \beta_0).$$

By our discussion of U-Statistics given above we have

$$N\mathbb{E} \left[(U_N^*(\beta_0) - U_N(\beta_0))^2 \right] \rightarrow 0$$

as $N \rightarrow \infty$ and hence, applying a Slutsky Theorem,

$$\boxed{\sqrt{N}(\hat{\beta} - \beta_0) \xrightarrow{D} \mathcal{N}(0, m^2 \Gamma_0^{-1} \Omega_0 \Gamma_0^{-1})}$$

U-Process Minimizers (continued)

To construct an estimate of the asymptotic variance of $\hat{\beta}$ we compute

$$\hat{h}_1(Z_i; \hat{\beta}) = \binom{N-1}{m-1}^{-1} \sum_{\mathbf{j} \in C_{m-1, N-1}} h(Z_i, Z_{j_2}, \dots, Z_{j_m}; \hat{\beta}),$$

and then calculate

$$\begin{aligned} \hat{\Omega} &= \frac{1}{N} \sum_{i=1}^N \hat{h}_1(Z_i; \hat{\beta}) \hat{h}_1(Z_i; \hat{\beta})', \\ \hat{\Gamma} &= \binom{N}{m}^{-1} \sum_{\mathbf{i} \in C_{m, N}} \nabla_{\beta\beta} l(Z_{i_1}, \dots, Z_{i_m}; \hat{\beta}). \end{aligned}$$

Application: partially linear logit

Consider the binary choice model

$$Y_i = 1 \left(X_i' \beta_0 + g(W_i) - U_i \geq 0 \right),$$

with U_i logistic.

Assume that W_i is discretely-valued, but perhaps with “many” support points.

An estimator which replaces the unknown function $g(W_i)$ with a vector of dummy variables for each support point of W_i may have poor finite sample properties and/or be difficult to compute.

Application: partially linear logit (continued)

Let i and j be two independent random draws.

Recalling results from binary choice with panel data analysis we have that

$$\begin{aligned} \Pr(Y_i = 0, Y_j = 1 | X_i, X_j, Y_i + Y_j = 1, W_i = W_j) \\ = \frac{\exp(X_j' \beta_0 + g(W_j))}{\exp(X_j' \beta_0 + g(W_j)) + \exp(X_i' \beta_0 + g(W_j))} \\ = \frac{\exp((X_j - X_i)' \beta_0)}{1 + \exp((X_j - X_i)' \beta_0)}. \end{aligned}$$

Application: partially linear logit (continued)

If we let

$$S_{ij} = \text{sgn} \{Y_j - Y_i\},$$

we may base estimation of β_0 on the U-Process

$$\begin{aligned} L_N(\beta) = & \binom{N}{2}^{-1} \sum_{i=1}^N \sum_{j < i} \left[\mathbf{1}(W_i = W_j) |S_{ij}| \right. \\ & \times \left. \left\{ S_{ij} (X_j - X_i)' \beta - \ln \left[1 + \exp \left(S_{ij} (X_j - X_i)' \beta \right) \right] \right\} \right]. \end{aligned}$$

Application: partially linear logit (continued)

To construct an estimate of the asymptotic variance of $\hat{\beta}$ first define

$$\begin{aligned} \hat{h}_1(Z_i; \hat{\beta}) = & \frac{1}{N-1} \sum_{j=1, j \neq i}^N \left[\mathbf{1}(Z_i = Z_j) |S_{ij}| \right. \\ & \times \left. \left\{ \mathbf{1}(S_{ij} = 1) - \frac{\exp\left((X_j - X_i)' \hat{\beta}\right)}{1 + \exp\left((X_j - X_i)' \hat{\beta}\right)} \right\} (X_j - X_i) \right]. \end{aligned}$$

Application: partially linear logit (continued)

We can then compute

$$\begin{aligned}\hat{\Gamma} &= -\frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \left[\mathbf{1}(Z_i = Z_j) |S_{ij}| \right. \\ &\quad \times \left. \left\{ \frac{\exp\left((X_j - X_i)' \hat{\beta}\right)}{\left[1 + \exp\left((X_j - X_i)' \hat{\beta}\right)\right]^2} \right\} (X_j - X_i) (X_j - X_i)'\right] \\ \hat{\Omega} &= \frac{1}{N} \sum_{i=1}^N \hat{h}_1(Z_i; \hat{\beta}) \hat{h}_1(Z_i; \hat{\beta})' .\end{aligned}$$

Wrapping Up

“Modern” applications may involve kernels which are indexed by N (e.g., semiparametric M-Estimation, Sparse Network Asymptotics).

Degeneracy: what happens if $\delta_1^2 = 0$? See Menzel (2021).

Two-sample U-Statistics (useful in Bi-Partite settings).

Exercises:

See Ferguson (2005) and Lehmann (1999) for many classic examples from nonparametric statistics which you can use as practice problems.

Compare exact distribution of sample variance in Gaussian model with U-Statistic large sample theory.

*References

Ferguson, T. S. (2005). U-statistics. University of California - Los Angeles.

Hoeffding, W. (1948). A class of statistics with asymptotically normal distribution. *Annals of Mathematical Statistics*, 19(3), 293 – 325.

Honoré, B. E. & Powell, J. L. (1994). Pairwise difference estimators of censored and truncated regression models. *Journal of Econometrics*, 64(1-2), 241 – 278.

Lehmann, E. L. (1999). *Elements of Large-Sample Theory*. New York: Springer.

Menzel, K. (2021). Bootstrap with cluster-dependence in two or more dimensions. *Econometrica*, 89(5), 2143 – 2188.

van der Vaart, A. W. (2000). *Asymptotic Statistics*. Cambridge: Cambridge University Press.